

Negative LDO (LT3093) — Working Document

Digital Audio Player · Rev 4

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1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LT3093EDD#TRPBF (package to confirm)
Package	12-lead DFN 3 × 3 mm, exposed pad (pin 13) = IN
Function	Ultralow-noise negative LDO, $-3V9 \rightarrow -3.3$ V analog rail

2. Design Requirements

Requirement	Value
Input	$-3V9$ from inverting charge pump (fed by $+3V9$), -3.9 V nominal
Output	-3.32 V
Load	I/V, summing, OPA1622 (V– side), DAC analog TBD
Output current	200 mA rated; ≈ 27 mA quiescent, ≈ 115 mA pk (32 Ω), ≈ 205 mA pk (16 Ω)
Current limit	Internal (ILIM to GND)
UVLO	-3.5 V rising
Power good	-3.0 V (90.4% of V_{OUT}), to MCU GPIO
VIOC	Unused — floated

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1, 2	IN	$-3V9$, C_IN
3	EN/UV	R_EN2 from IN, R_EN1 to GND
4	PG	R_PU to $+3V3$, MCU GPIO
5	PGFB	R_PG1 / R_PG2 midpoint
6	ILIM	GND
7	VIOC	No connect (float)
8	SET	R_SET $\parallel$ C_SET to GND
9	GND	GND
10	OUTS	OUT at load
11, 12	OUT	$-3V3A$, C_OUT, R_PG1
13	Exposed pad	IN ($-3V9$) — own pour and via field, never GND

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
R_SET	33.2 kΩ 0.1%	0402	YAGEO RT0402BRD0733K2L	#1-085
C_SET	2.2 μF (effective TBD)	25 V X7R, low leakage	Murata TBD	—
C_IN	10 μF (6.76 μF eff. at 3.9 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_OUT	10 μF (7.56 μF eff. at 3.3 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R_EN1	50 kΩ 0.1%	0603	YAGEO RT0603BRD0750KL	#1-075
R_EN2	88.7 kΩ 0.1%	0402	YAGEO RT0402BRD0788K7L	—
R_PG1	135 kΩ 0.1% (E192)	0402	YAGEO RT0402BRD07135KL	—
R_PG2	15 kΩ 0.1%	0402	YAGEO RT0402BRD0715KL	#1-078
R_PU	10 kΩ	0603	Generic	—

5. Equations and Results

Output voltage (range over I_{SET} 98 - 102 μA)

$$V_{OUT} = -I_{SET} R_{SET} = -100 \mu A \times 33.2 k\Omega = -3.32 V \quad (-3.254 \text{ to } -3.386 V)$$

Undervoltage lockout (I_{EN} term neglected — ignorable for $R_{EN1} < 100 k\Omega$)

$$V_{IN(UVLO)} = -1.26 \left(1 + \frac{R_{EN2}}{R_{EN1}} \right) - I_{EN} R_{EN2}$$

$$R_{EN2} = R_{EN1} \left(\frac{3.5}{1.26} - 1 \right) = 50 k\Omega \times 1.778 = 88.89 k\Omega \rightarrow 88.7 k\Omega$$

$$V_{IN(UVLO)} = -1.26 \left(1 + \frac{88.7}{50} \right) = -3.495 V \quad (-3.33 \text{ to } -3.69 V)$$

$$V_{IN(UVLO, falling)} = -(1.26 - 0.215) \left(1 + \frac{88.7}{50} \right) = -2.90 V \quad (\text{sim: } \approx -3.0 V)$$

Power good threshold (I_{PGFB} term neglected — ignorable for $R_{PG2} < 30 k\Omega$; R_PG1 is the top resistor)

$$V_{OUT(PG)} = -0.3 \left(1 + \frac{R_{PG1}}{R_{PG2}} \right) - I_{PGFB} R_{PG1}$$

$$R_{PG1} = R_{PG2} \left(\frac{3.0}{0.3} - 1 \right) = 15 k\Omega \times 9 = 135 k\Omega$$

$$V_{OUT(PG)} = -0.3 \left(1 + \frac{135}{15} \right) = -3.000 \text{ V (90.4\%)} \quad (-2.88 \text{ to } -3.12 \text{ V})$$

SET noise filter corner (1 μF effective)

$$f_c = \frac{1}{2\pi R_{SET} C_{SET}} = \frac{1}{2\pi \times 33.2 \text{ k}\Omega \times 1 \mu\text{F}} = 4.8 \text{ Hz}$$

Soft-start (fast start-up to PG threshold -3.0 V , then $R_{SET} \cdot C_{SET}$ tail)

$$\tau = R_{SET} C_{SET}, \quad t_{99} - t_{PG} = \tau \ln \frac{0.32}{0.033} = 2.27 \tau \approx 75 \text{ ms}/\mu\text{F} \quad (2.2 \mu\text{F} : 166 \text{ ms})$$

Headroom (sim dropout 214 mV at 201 mA; datasheet max 330 mV)

$$\Delta V_{PUMP(MAX)} = 3.9 - 3.287 - 0.33 = 0.28 \text{ V} \quad (0.12 \text{ V from the } -3.74 \text{ V step minimum})$$

Dissipation (205 mA pk, $I_{GND} \approx 6 \text{ mA}$)

$$P_D = 0.205 \times (3.981 - 3.32) + 0.006 \times 3.981 \approx 0.16 \text{ W}, \quad \Delta T_J \approx 0.16 \times 34 \approx 5.5^\circ\text{C}$$

Load current (1.41 V pk design maximum, both channels in phase)

Case	Per channel pk	Signal total	+ Quiescent	LT3093 pk
16 Ω full scale	88 mA	176 mA	27 mA	$\approx 205 \text{ mA}$
32 Ω full scale	44 mA	88 mA	27 mA	$\approx 115 \text{ mA}$
110 dB SPL, 32 Ω	7.9 mA	16 mA	27 mA	$\approx 43 \text{ mA}$

Dropout sweep (sim, ideal source; V_{OUT} 1% low = -3.287 V)

Load	V_{IN} at dropout	Dropout
29 Ω ($\approx 115 \text{ mA}$)	-3.490 V	203 mV
16.5 Ω ($\approx 201 \text{ mA}$)	-3.501 V	214 mV

C_{SET} sweep (sim, ideal -3.9 V source, 29 Ω load; model reads $\approx 0.70 \mu\text{V}$ vs datasheet 0.8 μV at 4.7 μF , 10 Hz - 100 kHz)

C_{SET} (effective)	t_{99} after PG	Noise, 20 Hz - 20 kHz
0.47 μF	$\approx 35 \text{ ms}$	1.44 $\mu\text{V RMS}$
1 μF	$\approx 75 \text{ ms}$	0.75 $\mu\text{V RMS}$
2.2 μF	166 ms (sim)	0.44 $\mu\text{V RMS}$
4.7 μF	$\approx 355 \text{ ms}$	0.34 $\mu\text{V RMS}$

Output impedance (sim, 29 Ω DC load)

Frequency	Z_OUT
1 kHz	0.26 m Ω
20 kHz	5.2 m Ω
282 kHz (peak)	76 m Ω

Startup and brownout (sim, 2.2 μ F C_SET)

Event	Rising (0 $\rightarrow$ -3.9 V in 20 ms)	Falling (-3.9 $\rightarrow$ 0 V in 50 ms)
Output starts / loses regulation	V_IN = -3.62 V ($\approx$ 0.6 ms after EN trip)	V_IN = -3.490 V
PG changes	V_OUT = -3.000 V, 3.9 ms after start	V_IN = -3.129 V
Settled 99% / shut down	166 ms after PG	V_IN = -2.957 V

Load step (sim, 27 $\leftrightarrow$ 205 mA, 1 μ s edges; identical at -3.9 V and -3.74 V input)

Item	Value
Peak deviation	-7.6 mV / +3.6 mV around -3.320 V
Ringing	Damped, $\approx$ 2 cycles at $\approx$ 140 kHz
Settling ($\pm$ 0.3 mV)	$\approx$ 20 μ s
DC shift 27 $\rightarrow$ 205 mA	Not visible ($<$ 0.1 mV)

Line transient and input ripple (sim, 205 mA load)

Item	Value
Line step -3.9 $\rightarrow$ -3.74 V (1 μ s edge)	61 μ V peak, settles $\approx$ 20 μ s ($\approx$ 68 dB)
Line regulation, 160 mV step	$\approx$ 2 μ V DC shift
Input ripple 7.7 mV pp at 1.2 MHz	1.67 μ V pp at OUT (PSRR predicts 0.98 μ V)

6. Important Values

Item	Value
SET pin current	100 μ A (98 - 102 μ A over temp)
EN/UV negative threshold	-1.26 V (-1.20 to -1.33 V), 215 mV hysteresis
PGFB threshold	-300 mV (288 - 312 mV), 7 mV hysteresis
Dropout	214 mV sim / 330 mV max at 200 mA
Allowed pump drop at 205 mA	0.28 V from -3.9 V; 0.12 V from -3.74 V (worst case)
Rated output current	200 mA
Internal current limit	400 mA typ, 220 mA min

Item	Value
GND pin current	2.35 mA quiescent; ≈ 7 mA at 200 mA (in dropout)
Shutdown current	3 μ A typ
UVLO divider current	≈ 28 μ A from $-3V_9$
PG divider current	≈ 22 μ A from OUT
Output noise	0.44 μ V RMS at 2.2 μ F, 20 Hz – 20 kHz (sim; model $\approx 15\%$ optimistic)
PSRR	-77.9 dB at 1.2 MHz, both -3.9 V and -3.74 V input (sim)
Output impedance	5.2 m Ω at 20 kHz; 76 m Ω peak at 282 kHz (sim)
Load step 27 $\leftrightarrow$ 205 mA	$-7.6 / +3.6$ mV, settles in ≈ 20 μ s; no dropout at -3.74 V (sim)
Line step / ripple	61 μ V for 160 mV step; 1.67 μ V pp from 7.7 mV pp at 1.2 MHz (sim)
Fast start-up	0 $\rightarrow -3.0$ V in 3.9 ms
Firmware delay	Wait ≥ 170 ms after PG before OPA1622 enable (2.2 μ F)
PG at cold start	Reads high until IN passes ≈ -1.8 V — ignore PG until $+3V_3$ is up
Brownout order	Dropout (-3.49 V) $\rightarrow$ PG low (-3.13 V) $\rightarrow$ off (-2.96 V)
C_OUT requirement	≥ 4.7 μ F, ESR < 30 m Ω , ESL < 1.5 nH
θ_{JA} (DFN)	34 $^{\circ}$ C/W
PG in shutdown	Pulled low — reads not-good (opposite of LT3042)
VIOC	Floated; ≈ 120 k Ω internal to GND

7. Open Problems

#	Problem
1	Charge pump not selected — drop at 205 mA must be < 0.12 V; between -3.49 and -3.13 V the rail is out of regulation with PG still good
2	Worst-case UVLO (-3.69 V) leaves ≈ 0.05 V to the -3.74 V step minimum before pump drop
3	16 Ω full-scale peak (≈ 205 mA) exceeds 200 mA rating, near 220 mA min current limit
4	Both rails at 16 Ω peak ≈ 410 mA from $+3V_9$ vs LTC3440 250 mA design current
5	Exposed pad is IN ($-3V_9$) — dedicated pour and via field; footprint must not tie to GND
6	Quiescent (27 mA) unconfirmed — I/V estimate, OPA1622 I_Q at ± 3.3 V unread
7	DAC analog supply allocation to this rail open
8	C_SET MPN and effective value (SimSurfing); ceramic SET cap is piezoelectric — tap test at Rev A or film
9	Confirm package (DFN vs MSOP-12E) and R_EN2 / R_PG1 MPNs
10	PG brownout warning is 2.2 ms at sim ramp rate — real brownout faster; consider hardware mute from PG
11	PG low specified at 100 μ A; 10 k Ω to $+3V_3$ sinks 330 μ A — verify or raise R_PU
12	Sims remaining: current limit, PG in shutdown, full chain with pump

8. Revision Log

Rev	Date	Change
0	2026-09-22	Initial issue: R_SET, UVLO, PG, soft-start, headroom, dissipation, open problems.
1	2026-09-22	PG divider to 135 k Ω / 15 k Ω , threshold –3.0 V; R_PG2 reuses #1-078.
2	2026-09-22	Sim results added: dropout, PSRR, noise sweep, Z_OUT, startup, brownout. C_SET 2.2 μ F 25 V. Closed: noise validation, startup knee, UVLO falling behaviour, firmware delay.
3	2026-09-22	Load step result added; closed from remaining sims.
4	2026-09-22	Line transient and 1.2 MHz ripple results added.